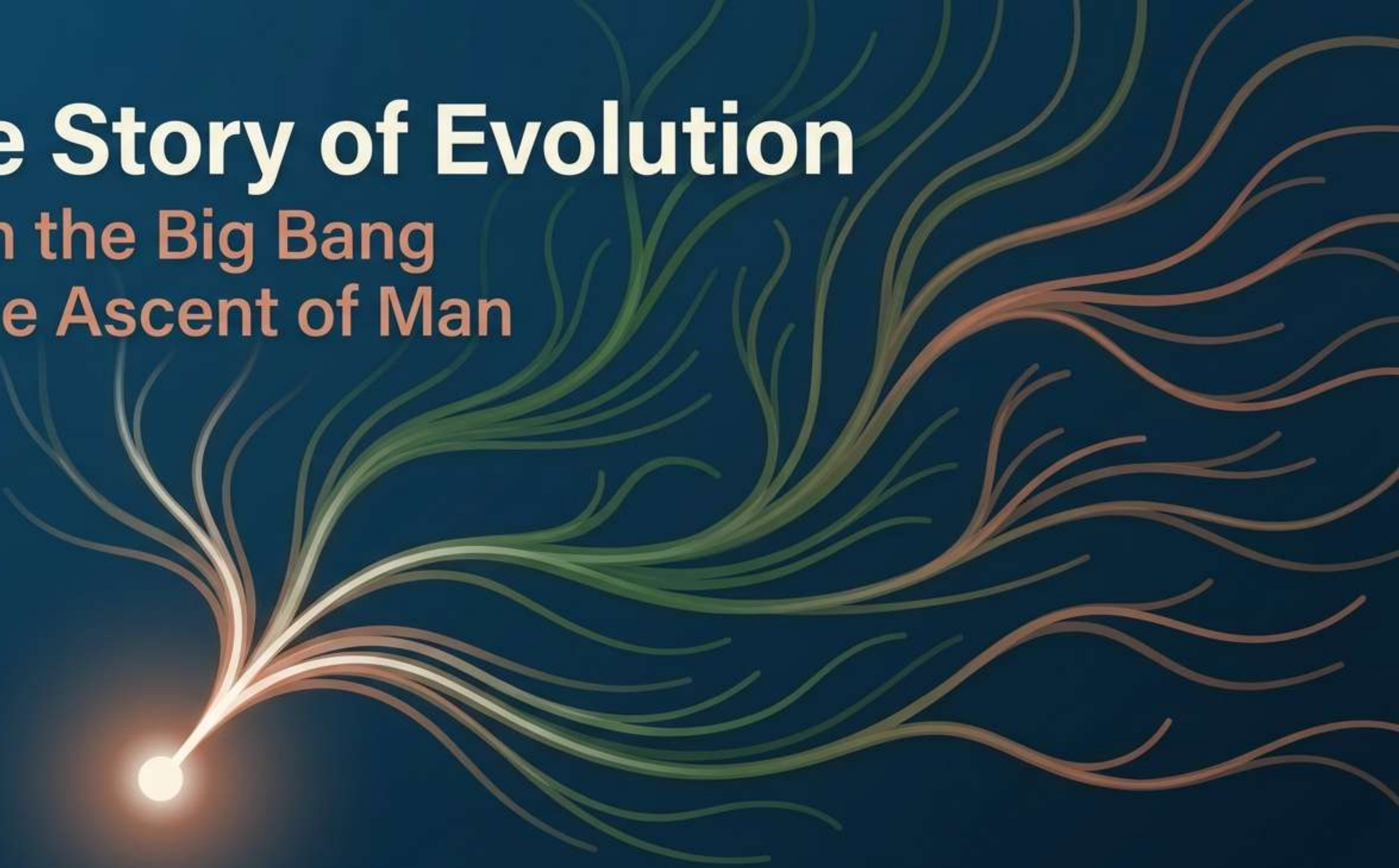


The Story of Evolution

From the Big Bang
to the Ascent of Man



13.8 Billion Years Ago (BYA)

The Big Bang creates an expanding, cooling universe, eventually forming galaxies like the Milky Way.

4.5 BYA

Earth forms. The primitive atmosphere is harsh and highly reducing—devoid of free oxygen. The surface is covered in water vapor, methane (CH_4), carbon dioxide (CO_2), and ammonia (NH_3) released from a molten mass.

4.0 BYA

(500 Million Years post-formation)

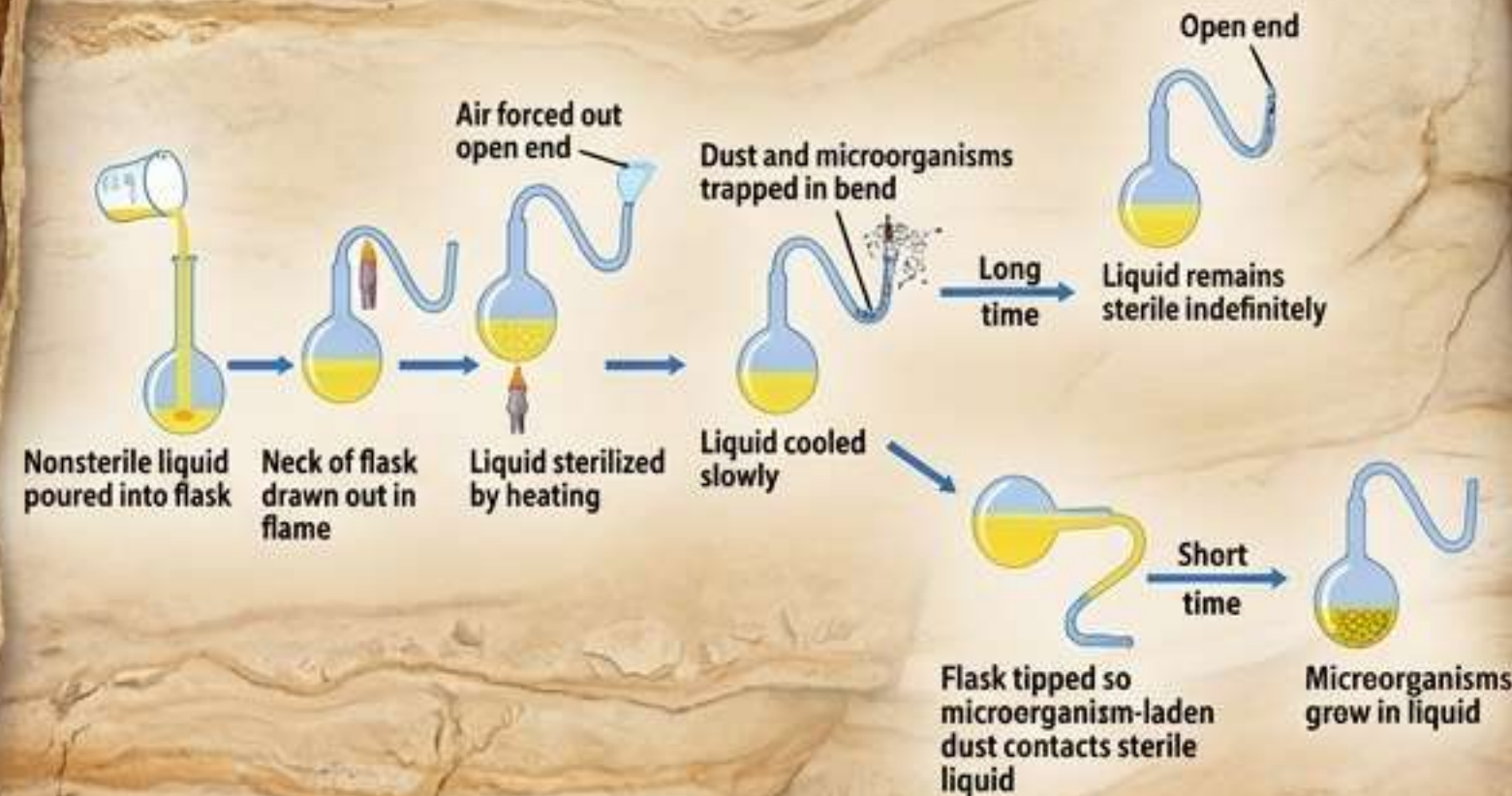
The Earth cools, water vapor falls as rain to form the first oceans, and the first forms of life appear.

Panspermia

Early Greek thought. The belief that units of life (“spores”) were transferred to Earth from outer space via meteorites.

Spontaneous Generation

The discarded belief that life arose from decaying matter (mud, straw). Louis Pasteur definitively disproved this using pre-sterilized swan-neck flasks, showing life only comes from pre-existing life (Biogenesis).

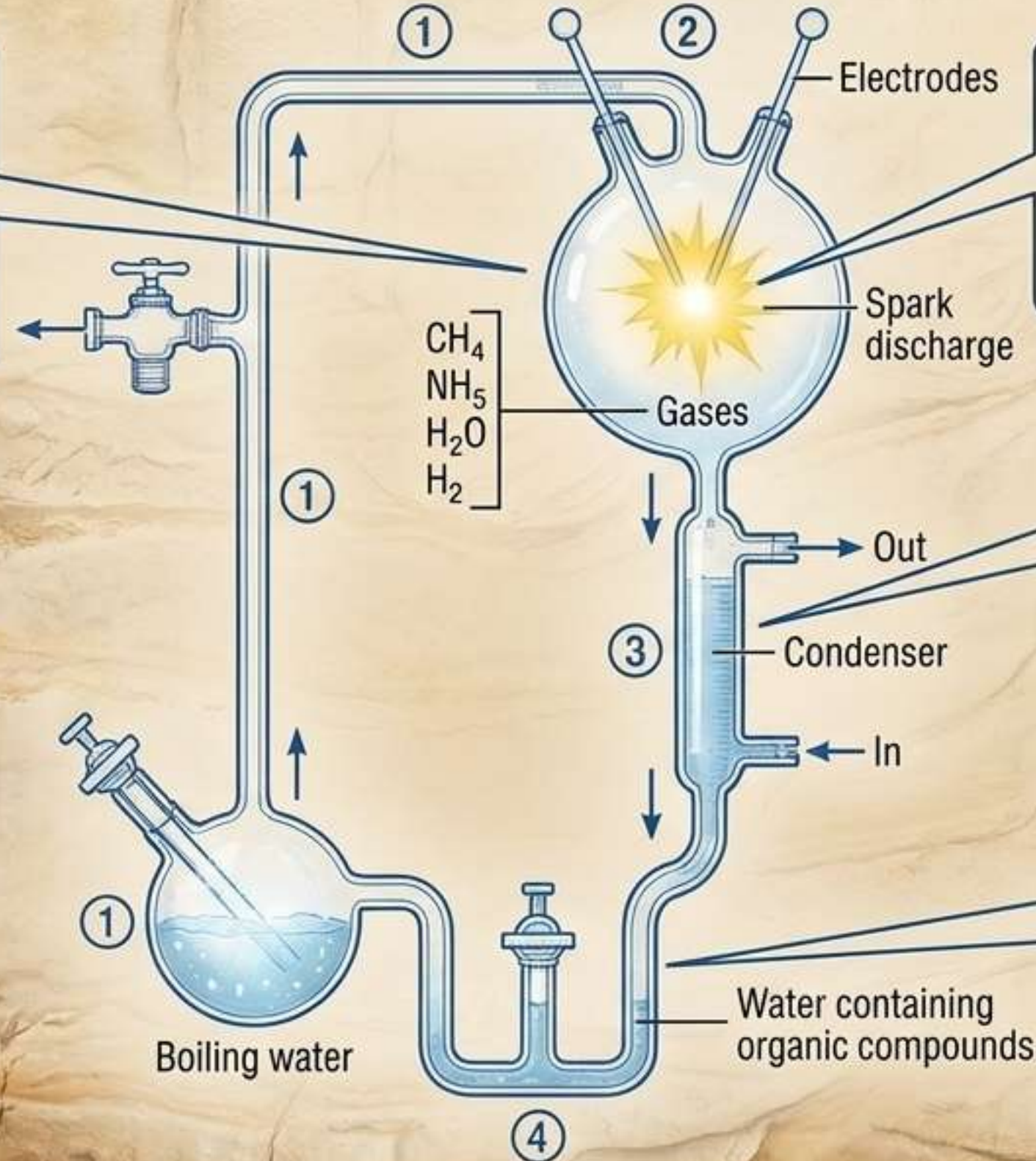


Chemical Evolution

Proposed by Oparin and Haldane. The accepted model stating that the first life originated from pre-existing, non-living organic molecules (RNA, proteins) under the extreme conditions of early Earth.

Recreating the Spark of Life

Step 1 (The Flask): S.L. Miller (1953) simulates the primitive Earth atmosphere using a mixture of CH_4 , H_2 , NH_3 , and water vapor.



Step 2 (The Spark): An electric discharge is fired into the closed flask at 800°C , simulating ancient volcanic storms and lightning.

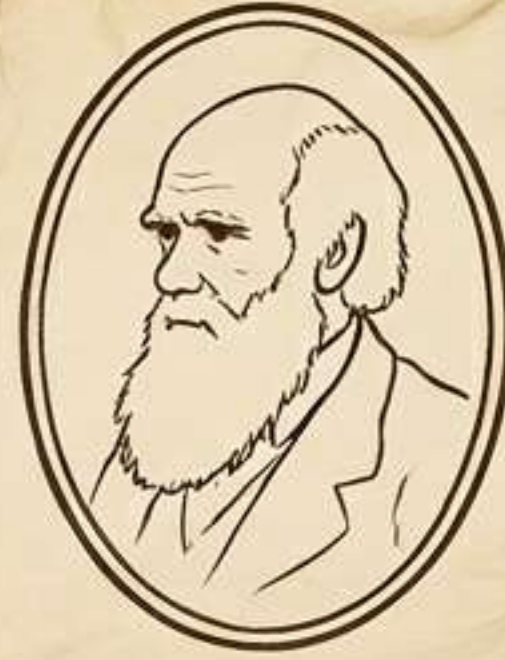
Step 3 (The Condenser): The mixture is cooled, mimicking primordial rain.

Result (The Trap): The resulting liquid contains newly synthesized amino acids—proving that inorganic constituents can form the organic building blocks of life.



Lamarck (Theory of Acquired Characters)

- French Naturalist. Proposed that evolution occurs through the 'use and disuse' of organs.
- Example: Giraffes stretched their necks to reach high leaves, and passed this acquired length to offspring. (Now largely discarded).



Darwin & Wallace (Natural Selection)

- British Naturalists. Based on HMS *Beagle* voyage observations.
- Proposed that populations have inherent variations. Individuals with traits best suited to the environment possess superior reproductive fitness. Nature 'selects' these survivors to propagate.

The Mechanism of Change: Variations vs. Mutations

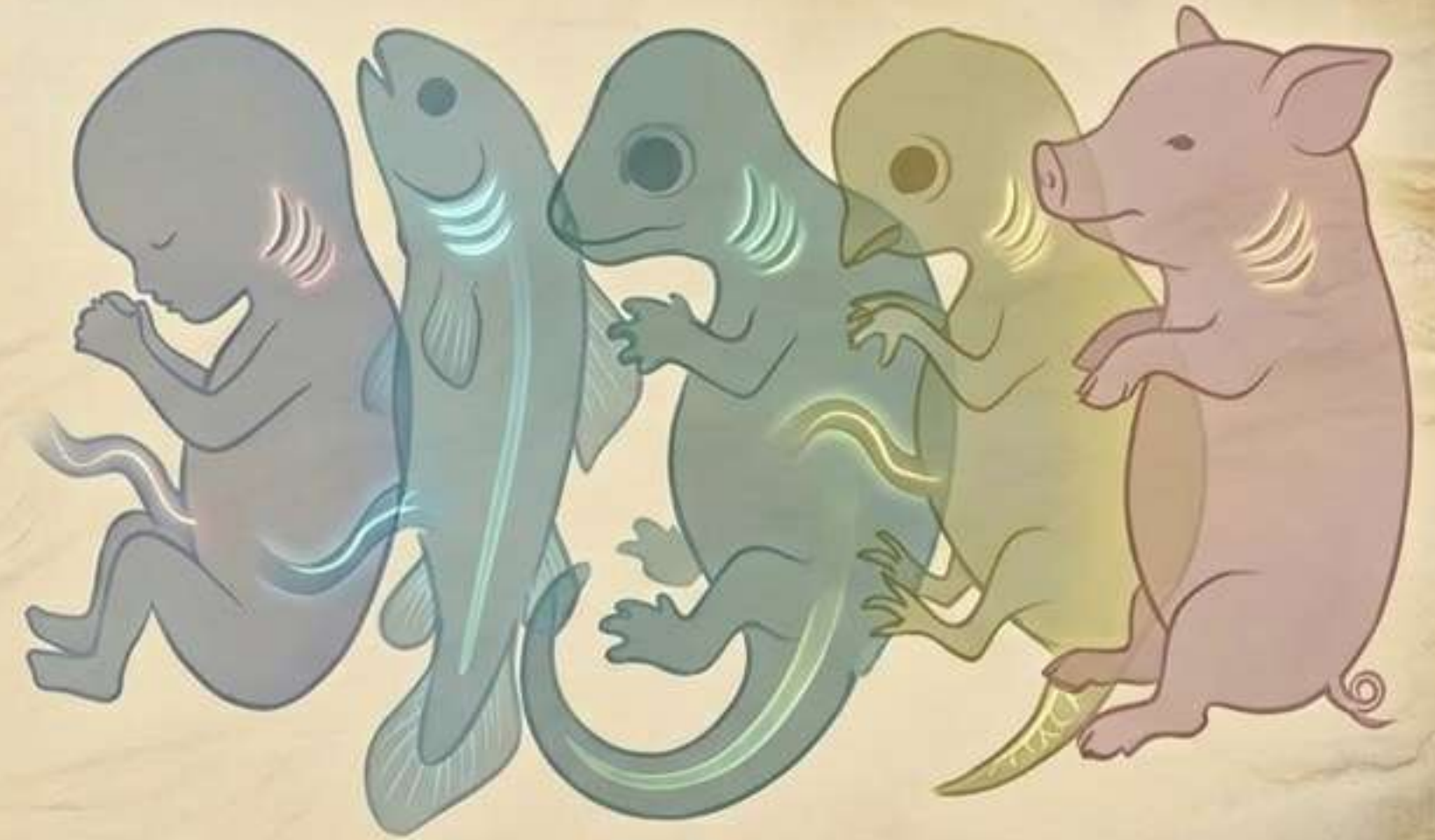
	Charles Darwin	Hugo de Vries
Origin of Theory	General natural observations.	Observations specifically on the evening primrose.
Nature of Change	Minor, slow, and gradual variations.	Large, abrupt, and sudden mutations.
Directionality	Directional changes leading slowly to speciation.	Directionless, random 'Saltation'—single-step large mutations creating immediate new species.

Paleontology



The Earth's geological history is a biological clock. Fossils (remains of hard parts) in different sedimentary rock strata correspond to the specific geological periods they lived in, revealing a chronological emergence of increasingly complex life (and the extinction of others, like dinosaurs).

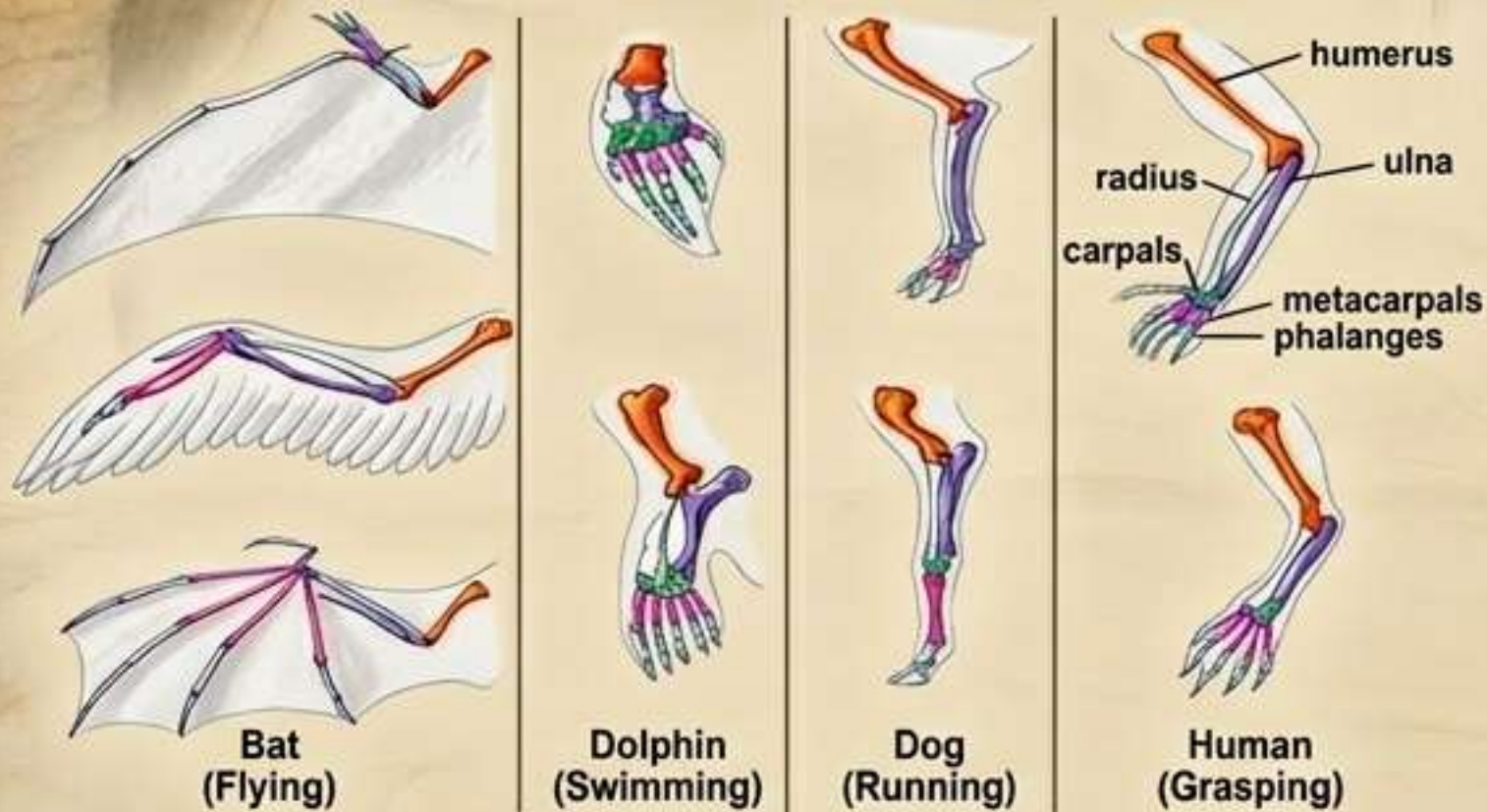
Embryology



Ernst Haeckel noted that all vertebrate embryos (including humans) temporarily develop vestigial gill slits, hinting at aquatic ancestors. (Note: Karl Ernst von Baer later refined this, showing embryos do not pass through the adult stages of other animals).

Structural Clues: Diverging and Converging Paths

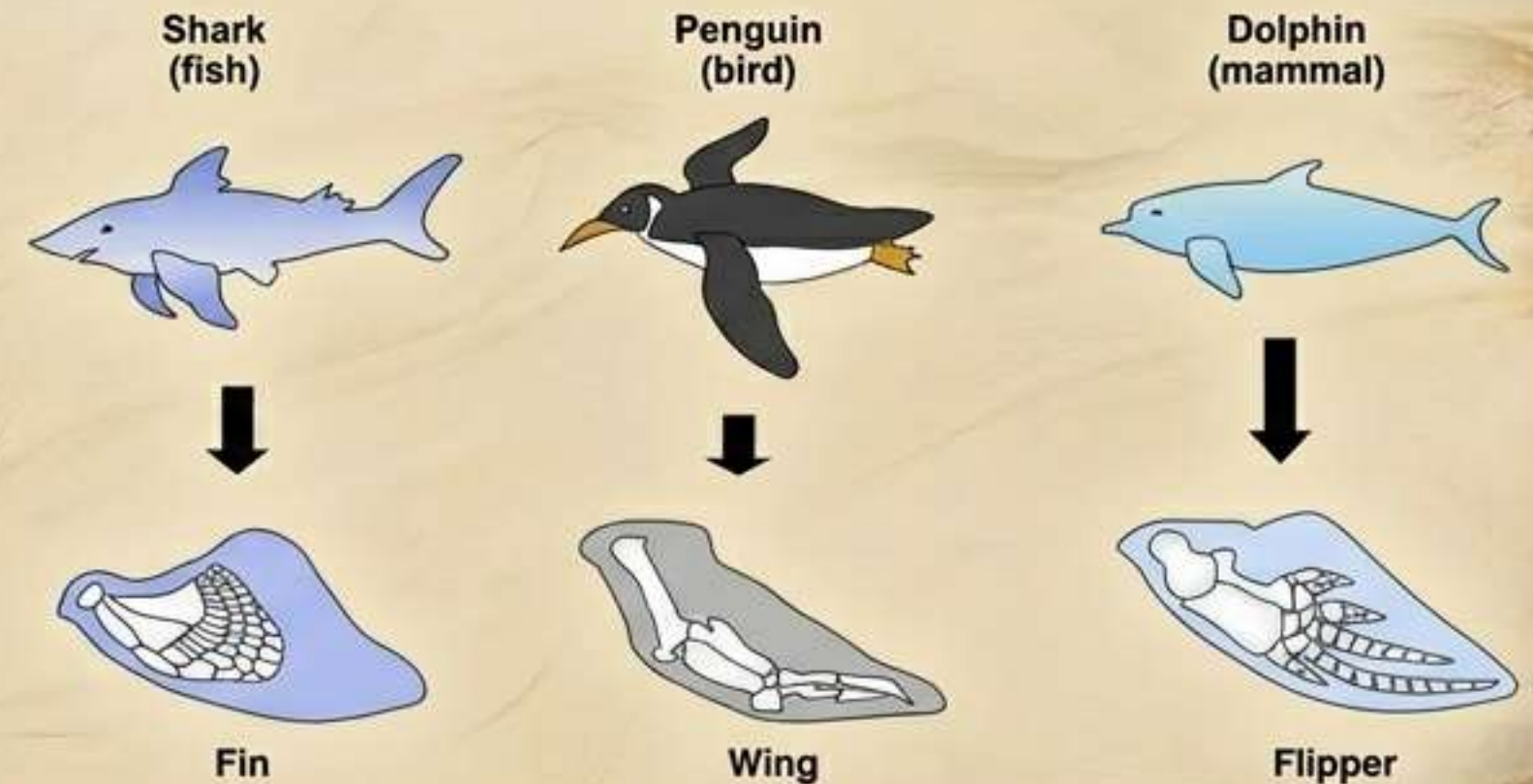
Homologous Organs (Divergent Evolution)



Same anatomical structure and origin, different functions. Developed along different directions due to varied needs.

Examples: Forelimbs of humans, cheetahs, whales, and bats. Bougainvillea thorns vs. *Cucurbita* tendrils. Indicates common ancestry.

Analogous Organs (Convergent Evolution)



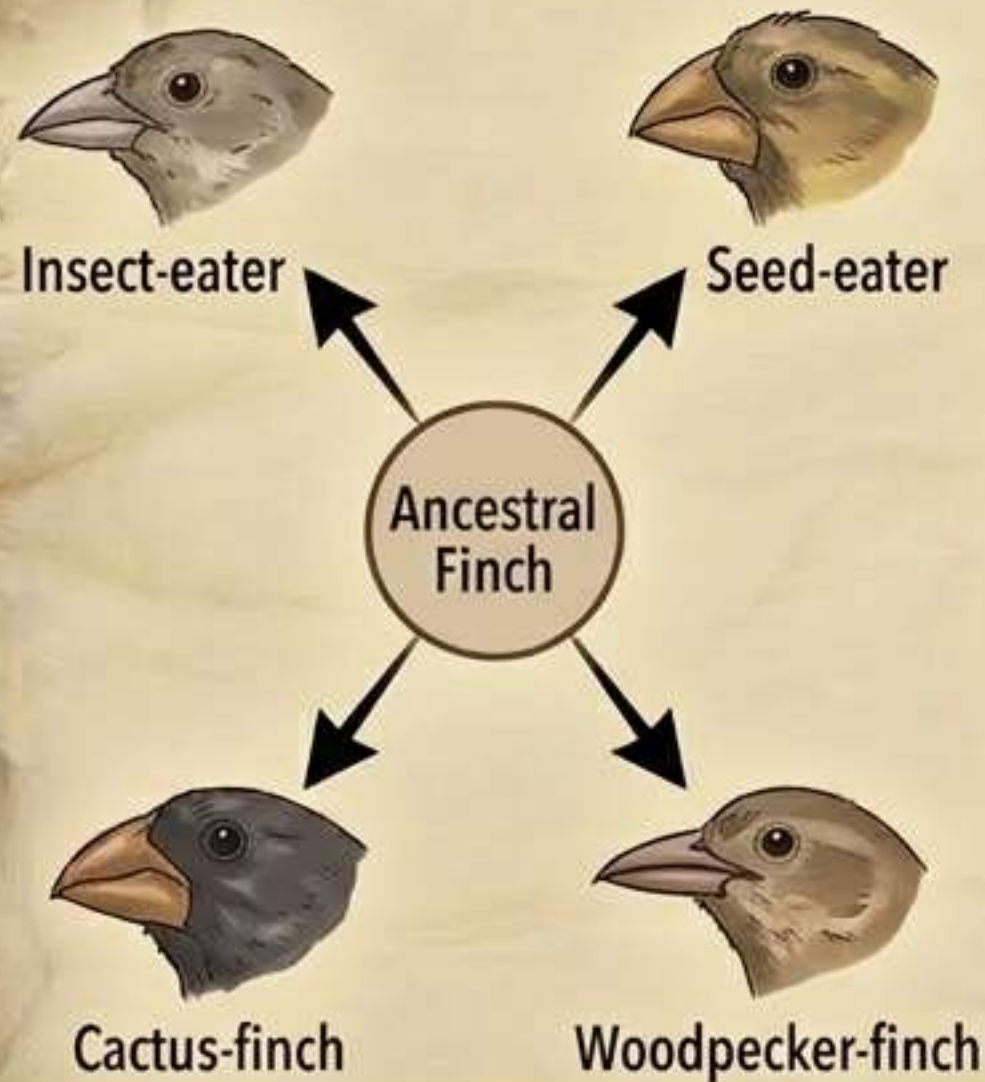
Different anatomical origins, similar functions. Shaped by similar environmental pressures.

Examples: Wings of butterflies vs. birds. Flippers of penguins vs. dolphins. Sweet potato (root) vs. potato (stem).

Adaptive Radiation

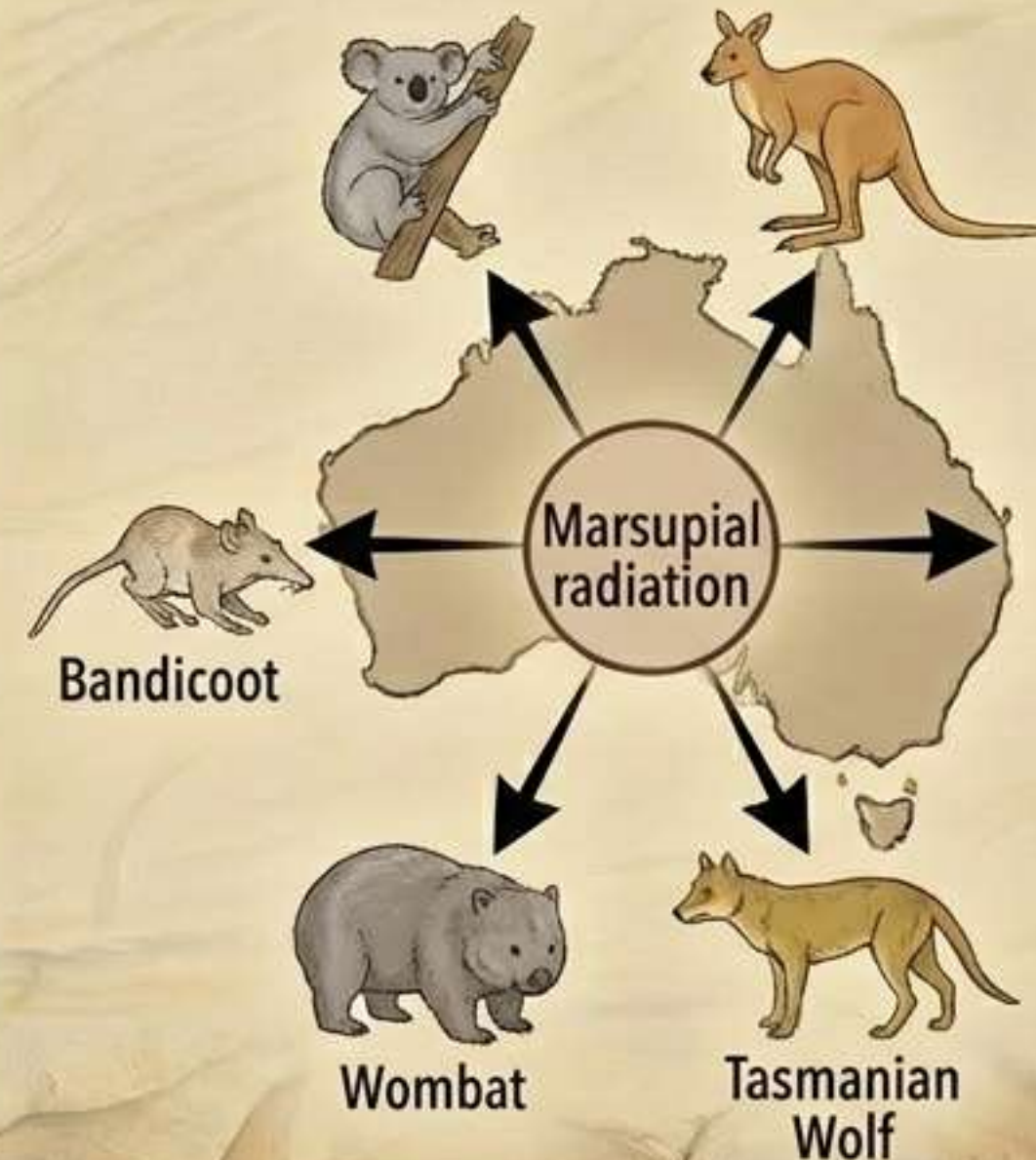
The evolutionary process where different species evolve in a given geographical area, starting from a single ancestral point and radiating into different habitats and ecological niches.

The Finches



A seed-eating ancestor on the Galapagos Islands radiated into insectivorous and vegetarian varieties with distinctly modified beaks.

The Marsupials



A single ancestral stock in isolated Australia radiated into diverse forms (koalas, kangaroos, Tasmanian wolves).

Natural Selection in Action: Industrial Melanism

1850 (Pre-Industrial England)



Tree trunks were covered in light-colored lichens. White-winged moths camouflaged perfectly, while dark-winged (melanised) moths were easily spotted and eaten by predators.

1920 (Post-Industrial England)



Pollution and soot killed the lichens and darkened the tree trunks. The survival advantage abruptly shifted: white moths were eaten, and the dark-winged moth population exploded.

Note: Modern anthropogenic (human-driven) evolution is visible in a fraction of the time through pesticide-resistant insects and antibiotic-resistant bacteria.

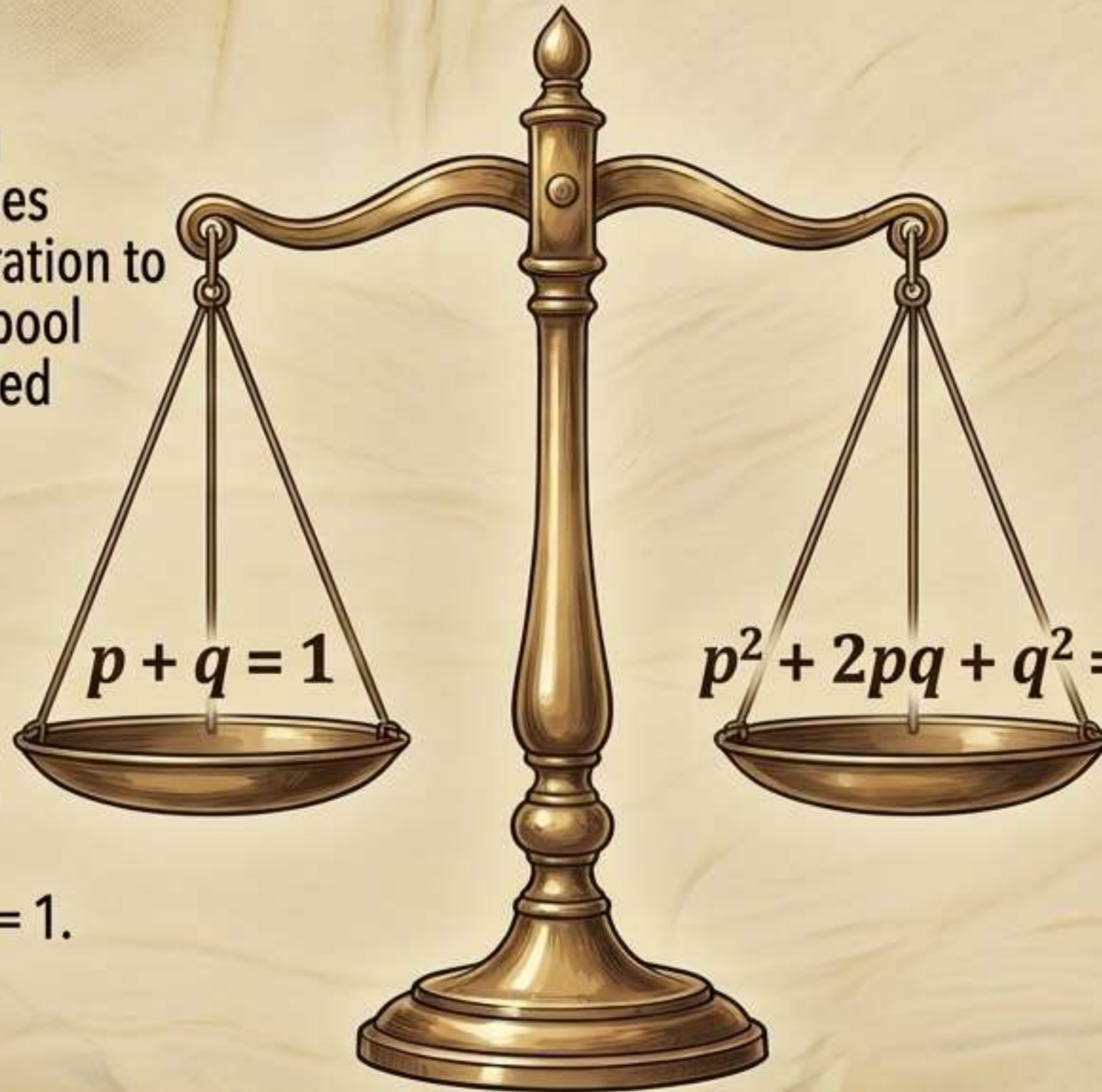
The Mathematics of Equilibrium

The Principle:

In a large, randomly mating population, allele frequencies remain constant from generation to generation. The total gene pool remains stable—a state called genetic equilibrium.

The Equation:

p^2 (homozygous dominant)
+ $2pq$ (heterozygous) +
 q^2 (homozygous recessive) = 1.



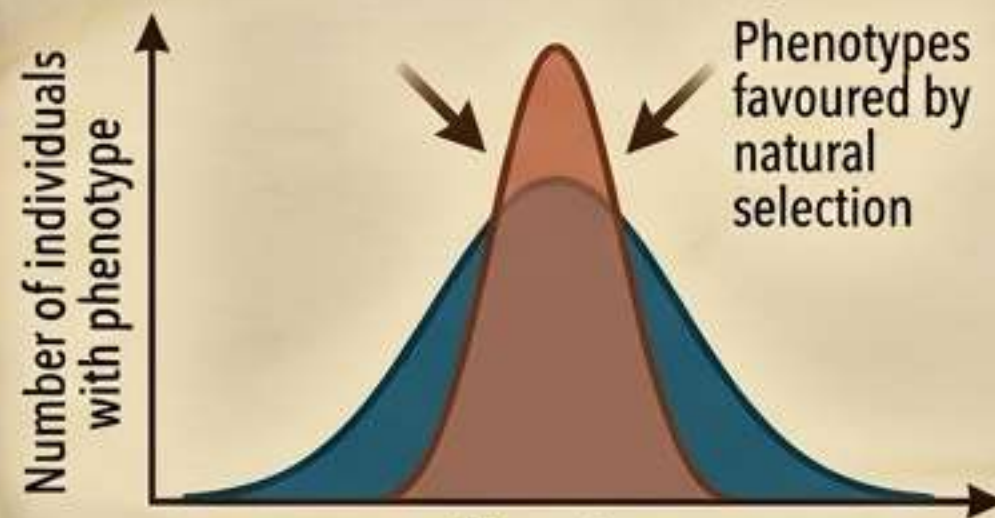
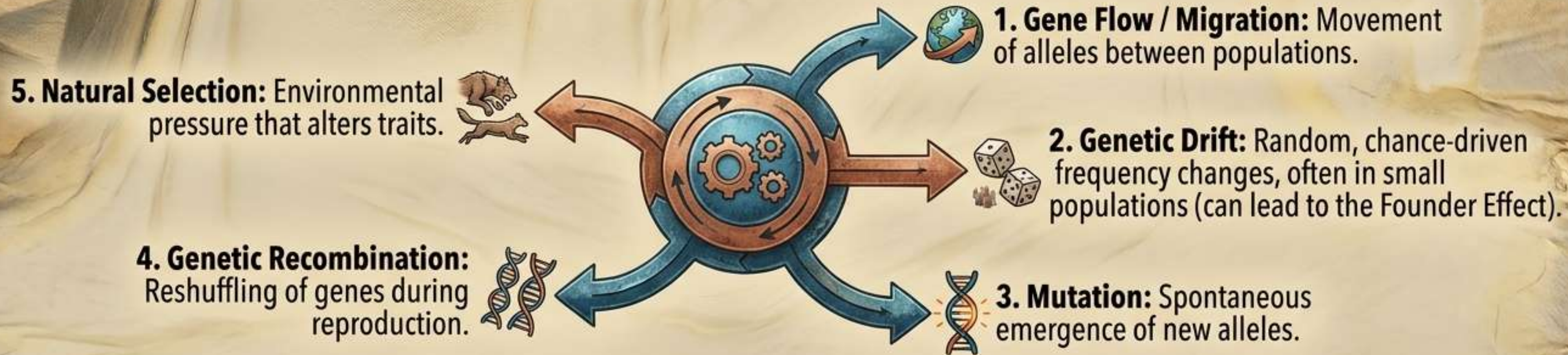
The Variables:

p represents the dominant allele frequency;
 q represents the recessive.

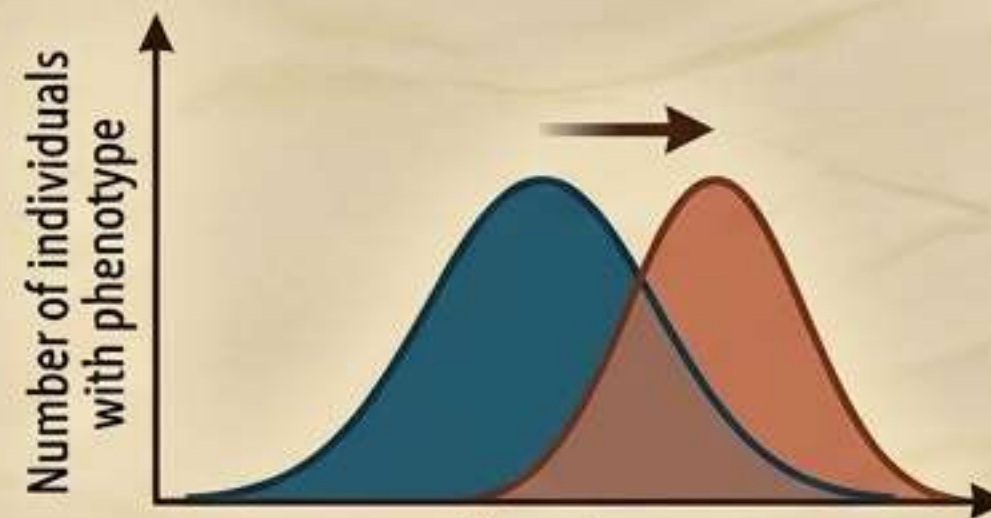
The Insight:

Any difference between expected Hardy-Weinberg values and observed reality indicates that evolutionary change is actively occurring.

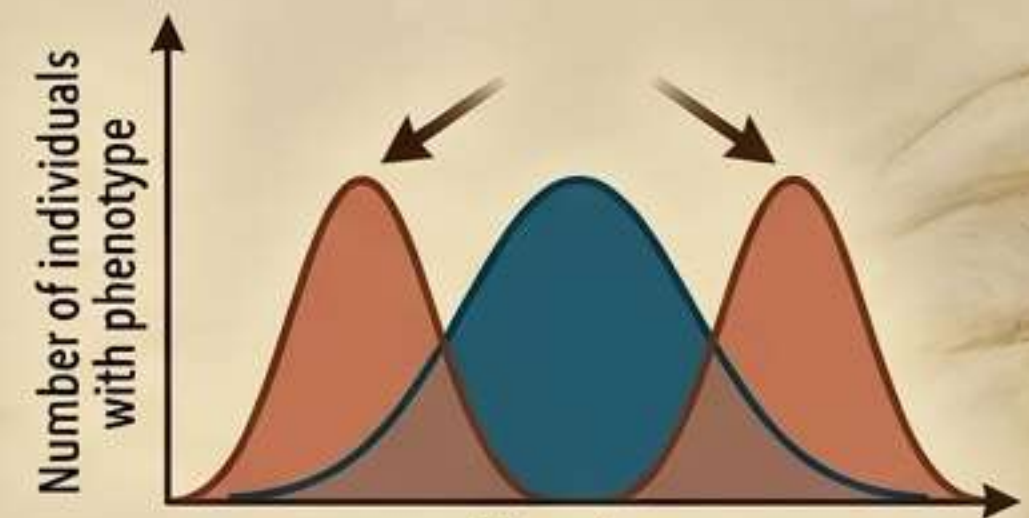
Five Forces That Disrupt the Balance



Stabilising
(favors the mean)



Directional
(favors one extreme)



Disruptive
(favors both extremes over the mean)

A Brief History of Flora and Fauna



2000 MYA

First single-celled life appears. Photosynthetic cells slowly oxygenate the atmosphere.



350 MYA

Emergence of jawless fish. Lobefin fish (like the Coelacanth) evolve into the first terrestrial amphibians.



Cenozoic Era

Small, shrew-like mammals survive, evolving viviparity (protecting unborn young internally) and eventually dominating the post-dinosaur landscape.



500 MYA

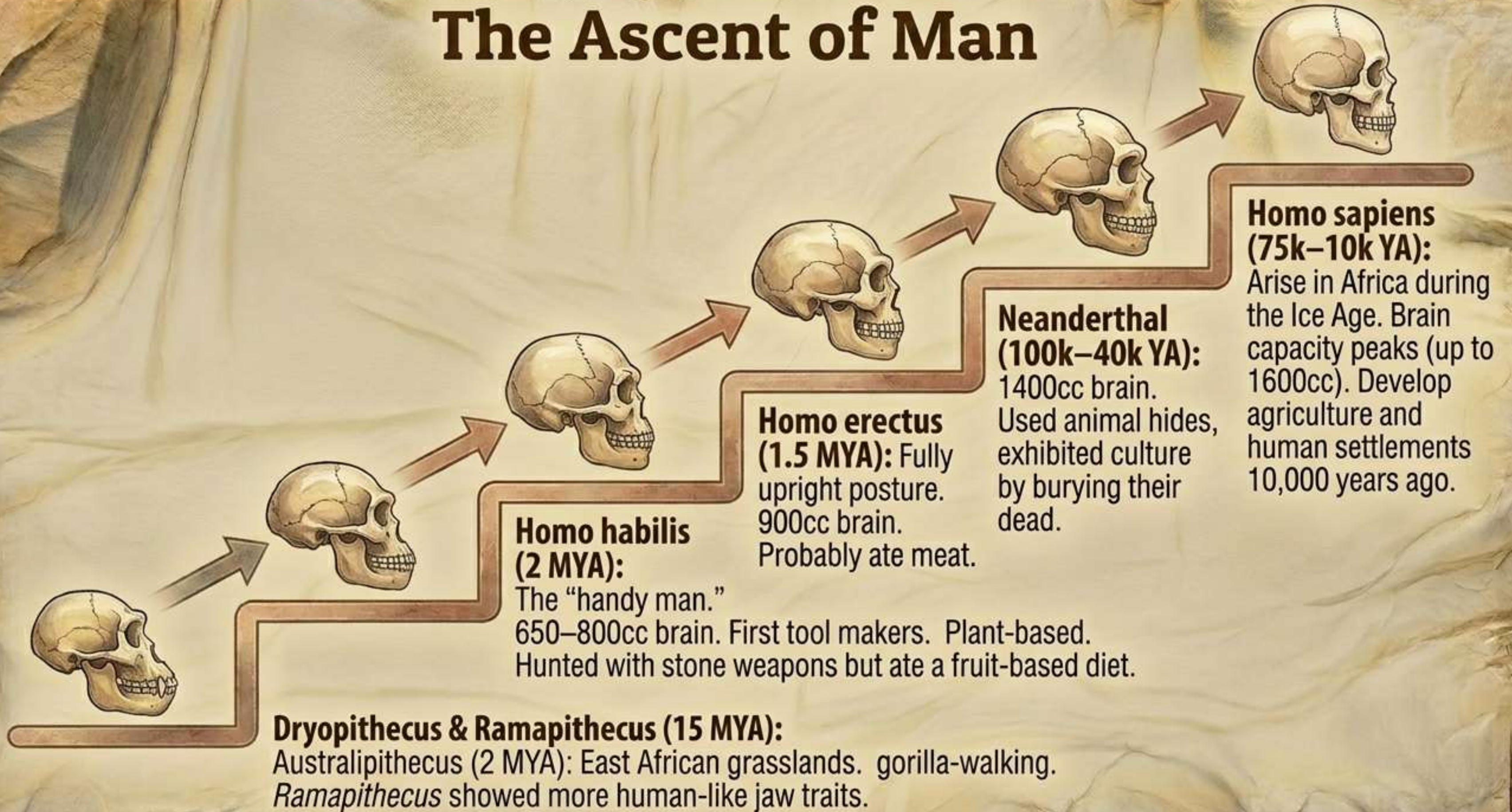
Invertebrates dominate the oceans.



Mesozoic Era

Amphibians evolve into reptiles, laying hard-shelled eggs. Dinosaurs (like the 20-foot *Tyrannosaurus rex*) rule the land until their sudden mass extinction 65 MYA.

The Ascent of Man



Synthesis: The Continuous Thread of Life

Evolution is not a predetermined ladder of progress. It is a stochastic, branching tree—driven by the blind chance of genetic mutation and the rigorous, unforgiving sieve of natural selection.

From the abiotic synthesis of amino acids in a primordial ocean to the mathematical equilibrium of modern genetics, the history of Earth and the biology of its inhabitants are fundamentally the same continuous story.

